

LLM-grounded Diffusion: Enhancing Prompt Understanding of Text-to-Image Diffusion Models with Large Language Models

Long Lian, Boyi Li, Adam Yala, Trevor Darrell

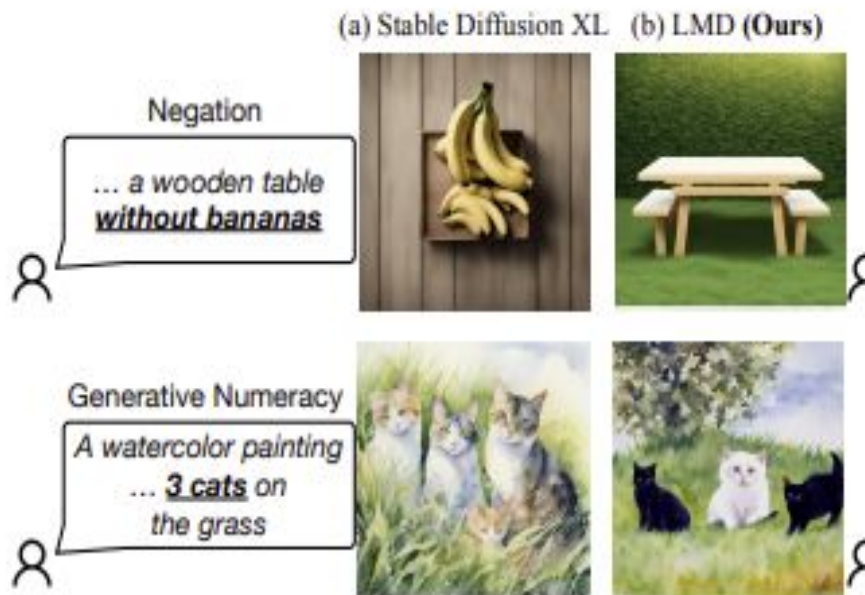
김수용, 강인하, 김태훈, 안가영, 안종식, 장보아

이미지 처리 2팀

Contents

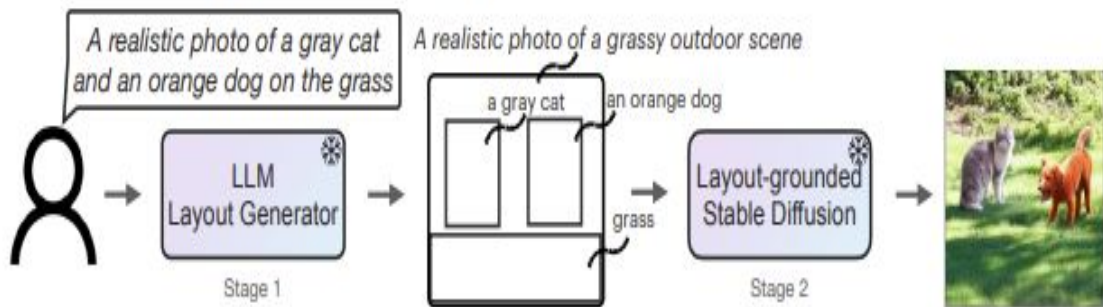
1. Introduction
2. Related Works
3. Method
4. Experiment
5. Ablation Study
6. Conclusion

Introduction - Problem



1. Diffusion struggle to accurately follow complex prompts
2. Unable to generate a certain number of objects or understand negation
3. Struggles with spatial reasoning or associating attributes
4. Gather huge multi-modal dataset requires considerable time

Introduction - LMD (Solution)



<LLM-grounded Diffusion>

1. **Training-Free** method that equips the diffusion model with an **LLM**
2. **Stage 1 : LLM** generates scene layouts in the form of captioned bounding boxes
3. **Stage 2 : Controller** guides an existing diffusion model without grounding in its training objective



Introduction - Contribution

1. **Training-free** two-stage generation pipeline that introduces **LLMs**
2. **Layout-grounded** Stable Diffusion, that generate images grounded on instance-level **box layouts** from the LLM
3. Enables **instruction-based scene specification** and allows **broader language support**
4. **Benchmark** to assess the prompt understanding ability of a text-to-image model

Contents

1. Introduction
2. Related Works
3. Method
4. Experiment
5. Ablation Study
6. Conclusion



Related Works

1. Text-to-image diffusion models

- Unsatisfactory performance when it comes to complex prompts
- Weak at binding attributes and spatial reasoning

2. LLMs for visual grounding

- Unable to generate layouts for objects not in the training set
- CLIP text embeddings to convey the information
=> Insufficient control
- Depends on a dataset annotated with boxes and captions



Related Works

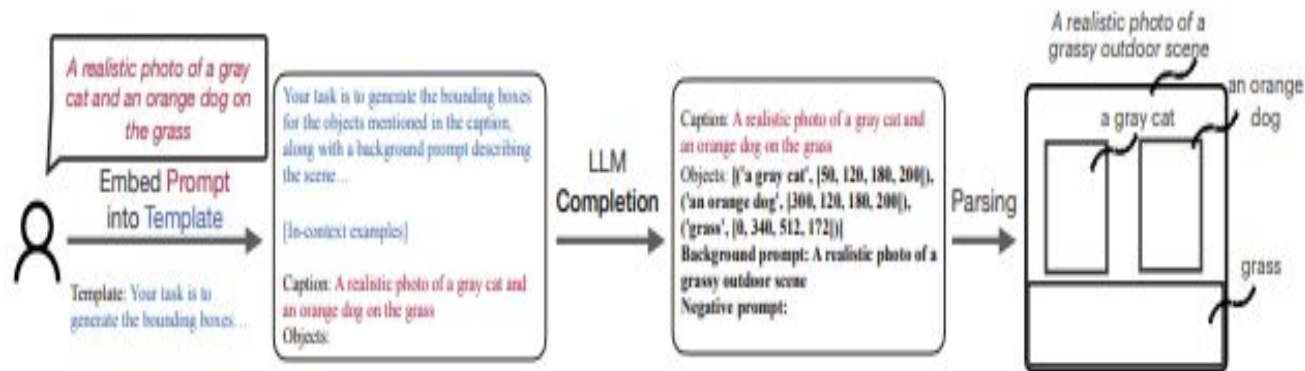
3. Spatially-conditioned image generation

- Rely on annotated external datasets with annotations such as boxes and captions.
- Training-based adaptation makes the model incompatible to add-ons (ex : LORA)

Contents

1. Introduction
2. Related Works
- 3. Method**
4. Experiment
5. Ablation Study
6. Conclusion

Method - Stage 1



1. LMD generates an image **layout** from a user prompt. LMD embeds the user prompt into a template with **instructions and in-context examples**.
2. LLM completion is parsed to obtain a set of captioned **bounding boxes**, a **background caption**, and an optional negative prompt.

Method - Stage 1

Instructions. Our text instructions to the LLM consist of two parts:

1. Task specification:

Your task is to generate the bounding boxes for the objects mentioned in the caption, along with a background prompt describing the scene.

2. Supporting details:

The images are of size 512×512... Each bounding box should be in the format of ... If needed, you can make reasonable guesses.

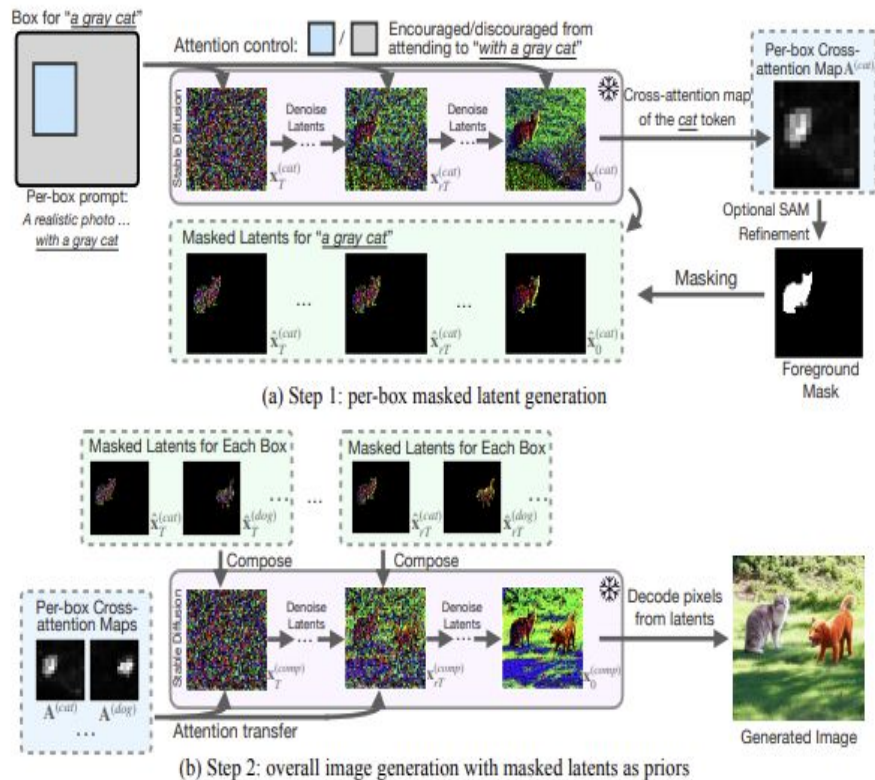
In-context learning. Similar to Brooks et al. (2023), we provide the LLM with manually curated examples after the task description. Through these examples, we clarify the layout representation and provide preferences to disperse ambiguity. An example is shown as follows:

Caption: A watercolor painting of a wooden table in the living room with an apple on it
Objects: [(‘a wooden table’, [65, 243, 344, 206]), (‘an apple’, [206, 306, 81, 69])]
Background prompt: A watercolor painting of a living room
Negative prompt:

LLM completion. After providing the in-context examples, we query the LLM for completion:

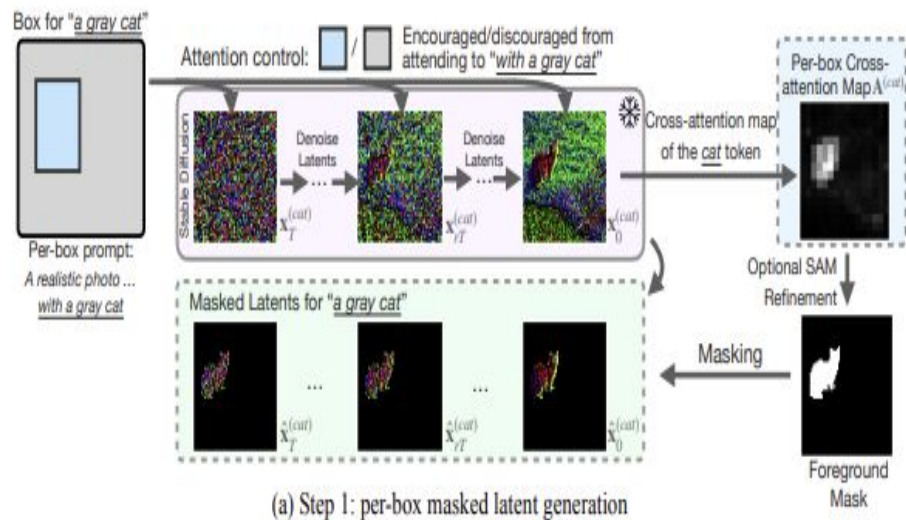
Caption: [input prompt from the user]
Objects: [start of LLM completion]

Method - Stage 2 (Overview)



1. Enables instance-level grounding by first generating **masked latents** for each individual **bounding box**
2. Composing the **masked latents** as priors to guide the overall image generation.

Method - Stage 2 (Part 1)

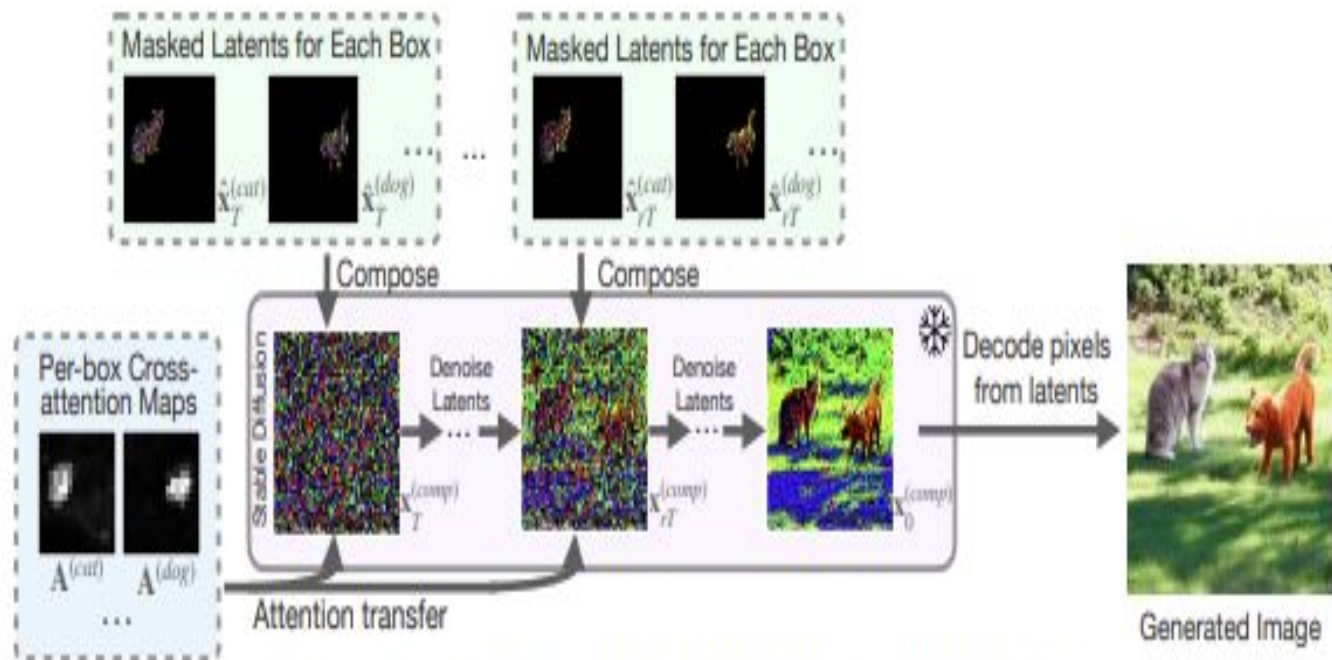


$$\mathbf{A}_{uv}^{(i)} = \text{Softmax}(\mathbf{q}_u^T \mathbf{k}_v)$$

$$E(\mathbf{A}^{(i)}, i, v) = -\text{Topk}_u(\mathbf{A}_{uv} \cdot \mathbf{b}^{(i)}) + \omega \text{Topk}_u(\mathbf{A}_{uv} \cdot (1 - \mathbf{b}^{(i)}))$$

1. Increase **Cross Attention** between Object (ex: gray cat) and **Box Area**
2. Decrease **Cross Attention** between Object and **Non Box Area**

Method - Stage 2 (Part 2)



(b) Step 2: overall image generation with masked latents as priors

Method - Additional Capabilities

Instruction-based Scene Specification



Initial user request

A realistic photo of a scene with a red car on the left.

LMD (Ours)



LMD (Ours)



Followup request 1

Add a blue car to the right



Followup request 2

Replace the car on the right with a yellow car

LMD (Ours)



LMD (Ours)



Followup request 3

Remove the yellow car. Instead, place flowers at that place

1. User can simply give clarifications or additional requests to the **LLM**
2. **LMD** remains consistent after multiple rounds of requests
3. **LMD** can handle requests that involve spatial reasoning

Contents

1. Introduction

2. Related Works

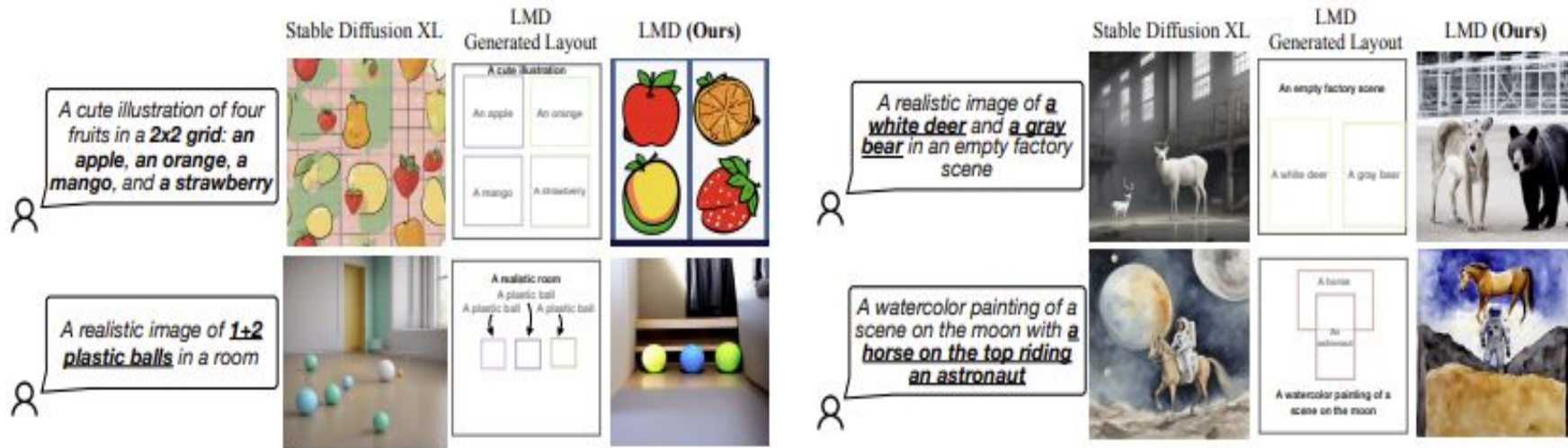
3. Method

4. Experiment

5. Ablation Study

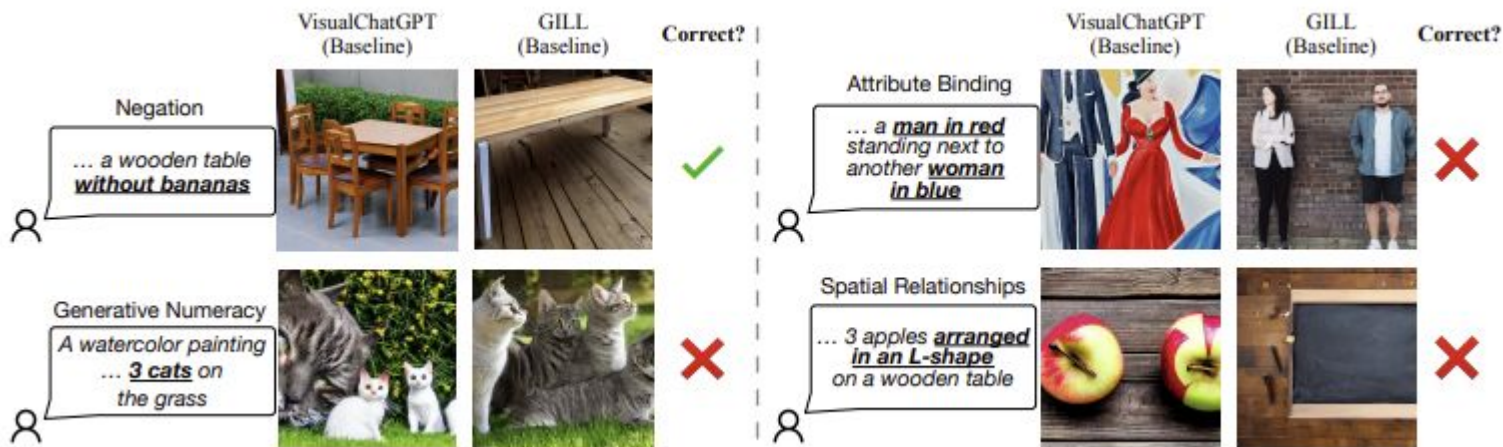
6. Conclusion

Experiment - Qualitative



1. Base Model : **SDXL**
2. Layout Generator : **gpt-4**
3. **LMD (ours)** greatly enhances prompt following ability

Experiment - Qualitative



1. **VisualChatGPT & GILL** still inherit the problems of insufficient control of text embeddings from the base SD model
2. **LMD** bypass the bottleneck of the text embedding representation that does not accurately convey spatial information

Experiment - Quantitative

Tasks	SD	Accuracy	
		LMD	LMD+
Negation	28%	100% (3.6×)	100% (3.6×)
Generative Numeracy	39%	62% (1.6×)	86% (2.2×)
Attribute Binding	52%	65% (1.3×)	69% (1.3×)
Spatial Relationships	28%	79% (2.8×)	67% (2.4×)
Average	37%	77% (2.1×)	81% (2.2×)

1. 100 prompts for each task
2. LLM : **GPT 3.5 Turbo**
3. LMD + : **GLIGEN**
4. Evaluator : **OWL-VIT**
5. Doubling the accuracy on average

Contents

1. Introduction

2. Related Works

3. Method

4. Experiment

5. Ablation Study

6. Conclusion

Ablation Study

Stage 1/Stage 2	Accuracy				
	Negation	Numeracy	Attribute	Spatial	Average
<i>Training-free methods:</i>					
LMD/MultiDiffusion (Bar-Tal et al., 2023)	100%	30%	42%	36%	52.0%
LMD/Backward Guidance (Chen et al., 2023)	100%	42%	36%	61%	59.8%
LMD/BoxDiff (Xie et al., 2023)	100%	32%	55%	62%	62.3%
LMD/LMD (Ours)	100%	62%	65%	79%	76.5% (+ 14.2)
<i>Training-based methods:</i>					
LMD/GLIGEN (Li et al., 2023b)	100%	57%	57%	45%	64.8%
LMD/LMD+ (Ours)	100%	86%	69%	67%	80.5% (+ 15.7)
LMD/LMD+ (Ours, GPT-4)	100%	84%	79%	82%	86.3% (+ 21.5)
<i>Evaluating generated layouts only (upper bound for image generation):</i>					
LMD/-	100%	97%	100%	99%	99.0%

Ablation Study

Method	Image Accuracy
	Average of 4 tasks
SD v1.5 (Default)	37%
LMD (on SDv1.5) (Ours , default)	77% (2.1×)
SD v2.1	38%
LMD (on SDv2.1) (Ours)	77% (2.0×)

Method	Image Accuracy
	Average of 4 tasks
LMD (w/o SAM)	72.8%
LMD (with SAM)	76.5%
LMD+ (w/o SAM)	82.8%
LMD+ (with SAM)	80.5%

1. Performance gains **2X** when switch from **SDv1.5** to **SDv2.1** without tuning hyperparameters
2. In **LMD**, removing **SAM** leads to a slight degradation
3. In **LMD+**, removing **SAM** slightly improves the results

Ablation Study

# shots	Layout Accuracy (4 tasks)	
	gpt-3.5-turbo	gpt-4
1 Shot	89.8%	100.0%
4 Shots	96.3%	100.0%
7 Shots	99.0%	100.0%

1. **GPT-3.5 (Default)** benefits from more in-context examples
2. **GPT-4** already excels in layout generation **even with only one example**
3. **GPT-4** is able to generate layouts without any in-context examples

Contents

1. Introduction
2. Related Works
3. Method
4. Experiment
5. Ablation Study
6. Conclusion

Conclusion



1. LLM may generate layouts that are **ambiguous** to the diffusion model
2. **Fine-tuning** the LLM to be more explicit about its assumptions in the layouts may alleviate **this problem**
3. **Additional request** for the side view and correct object sizes to the LLM



**THANK
YOU**